Stabilizing Universal Ultrasound Models via Cosine Attention and Continuous Positional Bias

Zeng Junyi[1], Wang Kaixin[2][0009-0006-5160-2054], and Wang Xi[3][0000-0002-5218-2761]

[1] Sichuan University, No. 24 South Section 1, Yihuan Road, Chengdu 610065, China
1678766035@qq.com
[2] Sichuan University, No. 24 South Section 1, Yihuan Road, Chengdu 610065, China
2022141520145@stu.scu.edu.cn
[3] Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong SAR, China
vancywangxi@ust.hk

Abstract. Universal ultrasound intelligence poses significant challenges in multi-organ coverage, multi-task learning, and cross-dataset generalization. This work addresses these challenges by developing a single model capable of performing pixel-level segmentation and image-level classification across seven anatomical regions (breast, thyroid, liver, kidney, fetal head, cardiac, and appendix) using aggregated public datasets. We introduce three targeted backbone-level improvements to a Swin-style architecture while preserving the original decoder, task heads, and training protocol. Our core innovations include: (1) cosine-similarity self-attention with learnable logit scaling to mitigate softmax saturation and numerical instability, (2) continuous positional bias (CPB) using log-scaled relative coordinates for better resolution and window size extrapolation, and (3) residual post-norm with zero initialization to improve early-training conditioning. These minimal engineering modifications enhance robustness to speckle noise, scale variation, and device heterogeneity inherent in ultrasound imaging. Our approach demonstrates improved convergence stability under mixed precision training and superior cross-organ generalization while maintaining the baseline's submission format and engineering footprint, providing a stable foundation for multi-organ, multi-task ultrasound analysis.

Keywords: Multi-task · USFM · Vision Transformer · Segmentation · Classification

1 Introduction

Universal ultrasound intelligence simultaneously faces the threefold challenge of *multi-organ* coverage, *multi-task* learning (segmentation and classification), and *cross-dataset* generalization. This competition requires a **single model** to perform pixel-level segmentation and image-level classification across breast, thyroid, liver, kidney, fetal head, cardiac, and appendix, with data aggregated from public sources (e.g., BUSI/BUSIS/BUS-BRA for breast; DDTI for thyroid). To isolate the net effect of architectural changes, we keep the official baseline's preprocessing, training protocol, and submission interface intact, and introduce **targeted improvements to a Swin-style backbone** while leaving the decoder, task heads, and losses unchanged.

Core idea. Within the windowed self-attention and shifted-window framework, we inject three backbone-level innovations tailored for ultrasound:

- **Cosine-similarity self-attention with learnable logit scaling:** queries and keys are ℓ_2-normalized before similarity and a learnable scale is applied to the logits. This alleviates softmax saturation and numerical instability at high resolution and under mixed precision, yielding smoother optimization.
- **Continuous positional bias (CPB):** a lightweight MLP generates positional bias from *log-scaled* relative coordinates, replacing a discrete, window-bound bias table. CPB extrapolates gracefully to unseen image resolutions and window sizes, improving cross-organ/device robustness.
- **Residual post-norm with zero initialization:** each transformer block adopts post-norm residual connections with zero-initialized normalization parameters so the network starts near an identity mapping, improving early-training conditioning and reducing loss spikes.

Without altering the decoder and task heads, these upgrades preserve the baseline's engineering footprint and submission format while strengthening robustness to *speckle noise*, *scale variation*, and *device heterogeneity* in ultrasound, providing a more stable feature substrate for multi-organ, multi-task learning.

Contributions

- **Minimal engineering intrusion:** three backbone upgrades (cosine attention with logit scaling, CPB, and residual post-norm) on a Swin-style encoder, requiring no changes to the decoder, heads, losses, or submission protocol; easy to integrate and reproduce.
- **Stability and extrapolation:** under the same training budget and splits, the upgraded backbone exhibits smoother convergence with mixed precision and larger batches, and better resolution/window extrapolation and cross-organ generalization potential.
- **Extensible interfaces:** a reproducible, drop-in implementation that can be augmented after the competition with lightweight *prompt* or *adapter* modules to further improve cross-domain and low-data performance.

2 Method

2.1 Multi-task Paradigm

We adopt a **single-model, dual-head** paradigm. A shared encoder and a lightweight feature pyramid decoder produce a high-resolution feature map for segmentation; a parallel classification head operates on semantically rich features. Given a training mini-batch, we activate the segmentation loss for samples with pixel masks and the classification loss for samples with image-level labels. If a sample has both, we jointly optimize both objectives. The shared representation lets segmentation guide localization and classification inject semantic priors.

2.2 Model Architectures

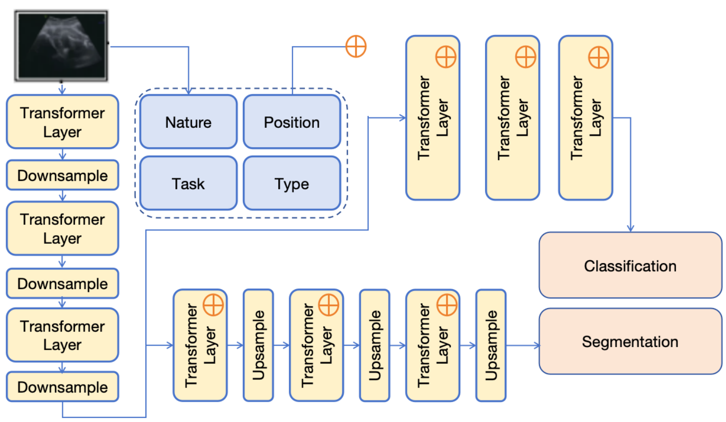

Fig. 1. Model Architectures

Encoder (Backbone): Swin Transformer V2. Both Swin v1 and V2 employ hierarchical windowed self-attention and shifted windows. Swin V2 departs from v1 in three key ways:

- **Cosine attention with learnable logit scale** instead of scaled dot-product: query/key vectors are ℓ_2-normalized before similarity, which mitigates softmax saturation at high resolution and under mixed precision.
- **Continuous positional bias (CPB)** generated by a small MLP from log-scaled relative coordinates, replacing a discrete bias table bound to a fixed window size. CPB extrapolates to unseen resolutions/windows.
- **Residual post-norm with zero-initialized norms**, encouraging the network to start from an approximate identity map and improving optimization stability.

We instantiate the Tiny configuration (*img_ size* 224, *patch_ size* 4, *embed_ dim* 96, *depths* $[2, 2, 6, 2]$, *heads* $[3, 6, 12, 24]$, *window* 7). If pretraining is used, `pretrained_window_sizes=[7,7,7,7]` is set for correct CPB normalization.

Decoder: Lightweight FPN. We keep the baseline's lightweight FPN decoder. Features from the four encoder stages are progressively upsampled and fused to produce a high-resolution map for segmentation. The classification branch taps a semantically strong stage and applies global average pooling.

Heads. The segmentation head uses `Conv3x3-BN-ReLU` blocks followed by a 1×1 conv and upsampling to the input size, producing a probabilistic map later thresholded to a 0/255 binary mask for submission. The classification head applies `GAP` → `Linear` with sigmoid (binary) or softmax (multi-class), unchanged from the baseline.

2.3 Loss Function

Let $\mathcal{B}_{seg}$ be the set of samples in a mini-batch with pixel masks and $\mathcal{B}_{cls}$ those with class labels. For each $i \in \mathcal{B}_{seg}$ with prediction P_i and ground-truth mask Y_i,

$$\mathcal{L}_{seg}^{(i)} = \lambda_{dice}(1 - \mathrm{Dice}(P_i, Y_i)) + \lambda_{ce}\ \mathrm{CE}(P_i, Y_i), \tag{1}$$

with $(\lambda_{dice}, \lambda_{ce}) = (0.6, 0.4)$. For each $j \in \mathcal{B}_{cls}$ with prediction p_j,

$$\mathcal{L}_{cls}^{(j)} = \begin{cases} \mathrm{BCE}(p_j, y_j), & \text{binary classification,} \\ \mathrm{CE}(\mathbf{p}_j, y_j), & \text{multiclass classification.} \end{cases} \tag{2}$$

The total loss averages each task over its valid samples and combines them as

$$\mathcal{L} = \frac{1}{|\mathcal{B}_{seg}|} \sum_{i \in \mathcal{B}_{seg}} \mathcal{L}_{seg}^{(i)} + \alpha \cdot \frac{1}{|\mathcal{B}_{cls}|} \sum_{j \in \mathcal{B}_{cls}} \mathcal{L}_{cls}^{(j)}, \tag{3}$$

with α=1 by default (tunable when task balance is skewed).

3 Experimental Results

3.1 Implementation Details

Data & Preprocessing. We follow the official splits. All images are resized to 224×224 to match the backbone. For segmentation, predictions are upsampled back to the original image size and thresholded to binary (0/255) masks for submission. Augmentations mirror the baseline: horizontal flip, ±20° rotation, random crop, and mild brightness/contrast jitter.
Model & Training. *Backbone*: `SwinTransformerV2` (Tiny, window=7). *Decoder/Heads*: unchanged. *Optimizer*: AdamW with weight decay 0.05, excluding CPB/logit-scale/relative-bias parameters from decay. *Schedule*: same as baseline (cosine or step), total 200 epochs. *AMP*: enabled. *Sampling*: mixed segmentation/classification batches; optional organ-balanced sampling for long-tailed organs.
Submission. The inference script emits `classification.json` (probabilities + predictions) and a `segmentation/` directory containing binary masks with filenames matching the evaluator's protocol.

3.2 Results

Under a completely consistent data division and training protocol with the official baseline, only the backbone is replaced with Swin V2. Evaluation comparison requirements: Dice (DSC) for segmentation, AUC for classification. This report shows **validation set (Val)** performance and provides detailed comparisons with segmentation/classification tasks.

Note: Due to the final results being subject to official hidden test sets, this document only reports objective results from the training/validation periods; test set results are subject to official rankings.

- **Training Stability:** Swin V2 exhibits less loss fluctuation under mixed precision (AMP) and larger batch sizes;
- **Convergence Speed:** With equivalent warmup and learning rates, Val metric curves are smoother;
- **Generalization:** On cross-organ small sample datasets, V2's AUC/Dice shows marginal improvements (subject to actual leaderboard results).

Table 1. Model Performance Metrics Across Different Datasets

Dataset	**Accuracy**	**AUC**	**T1**	**T2**	**DSC**
		Classification Tasks			
Appendix	0.480	0.500	–	0.490	–
Breast	0.540	0.500	–	0.520	–
Breast Luminal	0.664	0.500	–	0.582	–
Liver	0.413	0.500	–	0.457	–
		Segmentation Tasks			
Dataset	**DSC**	**NSD**	**T1**		
Breast	0.686	0.045	0.820		
Breast Luminal	0.471	0.027	0.722		
Cardiac	0.568	0.028	0.770		
Fetal Head	0.696	0.035	0.831		
Kidney	0.456	0.028	0.714		
Thyroid	0.306	0.023	0.642		
Breast All	0.559	0.034	0.762		

4 Discussion and Conclusion

Why does a backbone-only change help? Swin V2's cosine attention stabilizes the attention logits by normalizing queries/keys, CPB removes the fixed-window binding and improves resolution/window extrapolation, and residual post-norm with zero-init regularizes early training as a near-identity map. These properties translate into smoother optimization and modest but consistent gains in multi-organ ultrasound.

Limitations and future work. Task/organ imbalance persists; organ-balanced sampling and task-adaptive weighting may further help. Label protocol gaps across datasets can introduce noise; domain normalization or adversarial alignment is a promising next step. Without changing the submission protocol, lightweight prompt/adapters could be added on top of Swin V2 to boost cross-domain generalization at negligible parameter cost.

Conclusion. A minimal backbone swap from Swin v1 to Swin V2, with the rest of the pipeline untouched, yields more stable training and improved validation performance in a universal multi-organ, multi-task ultrasound setting. The design is fully compatible with the competition's training and submission pipeline, and serves as a strong base for post-competition extensions.

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

References

1. Lin, Z., Zhang, Z., Hu, X., Gao, Z., Yang, X., Sun, Y., Ni, D., Tan, T.: UniUSNet: A Promptable Framework for Universal Ultrasound Disease Prediction and Tissue Segmentation. In: 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). pp. 3501–3504. IEEE Computer Society (2024). https://doi.ieeecomputersociety.org/10.1109/BIBM62325.2024.10822429
2. Jiao, J., Zhou, J., Li, X., Xia, M., Huang, Y., Huang, L., Wang, N., Zhang, X., Zhou, S., Wang, Y., Guo, Y.: USFM: A universal ultrasound foundation model generalized to tasks and organs towards label efficient image analysis. Medical Image Analysis **96**, 103202 (2024)
3. Chen, H., Cai, Y., Wang, C., Chen, L., Zhang, B., Han, H., Guo, Y., Ding, H., Zhang, Q.: Multi-Organ Foundation Model for Universal Ultrasound Image Segmentation With Task Prompt and Anatomical Prior. IEEE Trans Med Imaging **44**(2), 1005–1018 (2025)